

Drugging the “Undruggable” MYCN Oncogenic Transcription Factor: Overcoming Previous Obstacles to Impact Childhood Cancers



Adam J. Wolpaw^{1,2}, Richard Bayliss³, Gabriele Büchel^{4,5}, Chi V. Dang^{2,6}, Martin Eilers⁴, W. Clay Gustafson⁷, Gwenn H. Hansen⁸, Natalia Jura^{9,10}, Stefan Knapp¹¹, Mark A. Lemmon¹², David Levens¹³, John M. Maris^{1,14}, Ronen Marmorstein¹⁴, Steven J. Metallo¹⁵, Julie R. Park¹⁶, Linda Z. Penn¹⁷, Michael Rape¹⁸, Martine F. Roussel¹⁹, Kevan M. Shokat²⁰, William P. Tansey²¹, Kliment A. Verba²², Seychelle M. Vos²³, William A. Weiss²⁴, Elmar Wolf²⁵, and Yaël P. Mossé^{1,14}

ABSTRACT

Effective treatment of pediatric solid tumors has been hampered by the predominance of currently “undruggable” driver transcription factors. Improving outcomes while decreasing the toxicity of treatment necessitates the development of novel agents that can directly inhibit or degrade these elusive targets. MYCN in pediatric neural-derived tumors, including neuroblastoma and medulloblastoma, is a paradigmatic example of this problem. Attempts to directly and specifically target MYCN have failed due to its similarity to MYC, the unstructured nature of MYC family proteins in their monomeric form, the lack of an understanding of MYCN-interacting proteins and ability to test

their relevance *in vivo*, the inability to obtain structural information on MYCN protein complexes, and the challenges of using traditional small molecules to inhibit protein–protein or protein–DNA interactions. However, there is now promise for directly targeting MYCN based on scientific and technological advances on all of these fronts. Here, we discuss prior challenges and the reasons for renewed optimism in directly targeting this “undruggable” transcription factor, which we hope will lead to improved outcomes for patients with pediatric cancer and create a framework for targeting driver oncoproteins regulating gene transcription.

MYCN as an Attractive Drug Target

Transcription factors in the MYC family are dysregulated in the majority of human tumors, including most pediatric malignancies (1, 2). This family is composed of three genes, *MYC* (*c-MYC*), *MYCN* (*n-MYC*), and *MYCL*, with conserved roles in central cellular processes, including regulating transcription, metabolism, and cell division. Whereas MYC is altered across a wide range of cancers, MYCN has a more narrow role—primarily as a driver of pediatric malignancies derived from central and peripheral nervous system

tissues, including neuroblastoma, medulloblastoma, retinoblastoma, astrocytoma, atypical teratoid/rhabdoid tumors (ATRT), and glioblastoma multiforme, among others, with emerging roles as a driver of therapy-resistant neuroendocrine variants of lung and prostate cancer (2).

MYCN is in many ways an ideal therapeutic target. Unlike MYC, its physiologic expression is tightly lineage restricted during development, with limited expression in normal pediatric or adult tissues, suggesting a wide therapeutic index for MYCN-specific drugs. When present in tumors, MYCN amplification is generally thought to be a

¹Division of Oncology and Center for Childhood Cancer Research, Children's Hospital of Philadelphia, Philadelphia, Pennsylvania. ²Wistar Institute, Philadelphia, Pennsylvania. ³Astbury Center for Structural Molecular Biology, School of Molecular and Cellular Biology, University of Leeds, Leeds, United Kingdom. ⁴Department of Biochemistry and Molecular Biology, Theodor Boveri Institute, Biocenter, University of Würzburg, Würzburg, Germany. ⁵Mildred Scheel Early Career Center, University Hospital Würzburg, Würzburg, Germany. ⁶Ludwig Institute for Cancer Research, New York, New York. ⁷University of California San Francisco, UCSF Benioff Children's Hospital, San Francisco, California. ⁸Nurix Therapeutics, San Francisco, California. ⁹Cardiovascular Research Institute, University of California San Francisco, San Francisco, California. ¹⁰Department of Cellular and Molecular Pharmacology, University of California San Francisco, San Francisco, California. ¹¹Institut für Pharmazeutische Chemie und Structural Genomics Consortium, Goethe-University Frankfurt, Frankfurt, Germany. ¹²Department of Pharmacology and Cancer Biology Institute, Yale School of Medicine, New Haven, Connecticut. ¹³Laboratory of Pathology, Center for Cancer Research, National Cancer Institute, Bethesda, Maryland. ¹⁴Perelman School of Medicine at the University of Pennsylvania, Philadelphia, Pennsylvania. ¹⁵Department of Chemistry, Georgetown University, Washington, DC. ¹⁶Department of Pediatrics, University of Washington School of Medicine and Center for Clinical and Translational Research, Seattle Children's Hospital, Seattle, Washington. ¹⁷Princess Margaret Cancer Centre, Toronto, Ontario, Canada. ¹⁸Howard Hughes Medical Institute, University of California at Berkeley, Berkeley, California. ¹⁹Department of Tumor Cell

Biology, St. Jude Children's Research Hospital, Memphis, Tennessee.

²⁰Howard Hughes Medical Institute and Department of Cellular and Molecular Pharmacology, University of California San Francisco, San Francisco, California. ²¹Vanderbilt University School of Medicine, Nashville, Tennessee. ²²Department of Pharmaceutical Chemistry, University of California San Francisco, San Francisco, California. ²³Department of Biology, Massachusetts Institute of Technology, Cambridge, Massachusetts. ²⁴Departments of Neurology and Pediatrics, Neurological Surgery and Helen Diller Family Comprehensive Cancer Center, University of California, San Francisco, California. ²⁵Cancer Systems Biology Group, Theodor Boveri Institute, University of Würzburg, Würzburg, Germany.

Note: To develop a novel discovery pipeline for the development of MYCN-targeting therapeutics, Alex's Lemonade Stand Foundation brought together a small group of multidisciplinary researchers from November 6 to 8, 2019, to discuss the challenges and opportunities to directly target the MYCN oncoprotein, which is an essential driver of many childhood cancers.

Corresponding Authors: Yaël P. Mossé, Pediatrics/Oncology, Children's Hospital of Philadelphia, Philadelphia, PA 19104. Phone: 215-590-0965; E-mail: mosse@chop.edu; and John M. Maris, maris@chop.edu
Cancer Res 2021;81:1627–32

doi: 10.1158/0008-5472.CAN-20-3108

©2021 American Association for Cancer Research.

truncal-initiating event that is required for ongoing tumor maintenance. It is rarely a subclonal finding and is not acquired or lost during tumor progression or relapse. In addition, transgenic expression of MYCN in the appropriate progenitor cells in mouse models can drive tumorigenesis that faithfully recapitulates human neuroblastoma and medulloblastoma, respectively (3, 4).

Despite intensive investigative efforts, indirectly targeting modulators of MYCN transcription and stability or the downstream mediators of MYCN function has failed to result in the identification of MYCN-specific therapeutics. For example, there was initial optimism that bromodomain and extra-terminal (BET) protein inhibitors could serve as universal and specific MYC-targeting drugs (5). However, it was clear early on that also affected MYCN (6) and MYCL (7). Preclinical data in neuroblastoma models showed tumor growth delay in some models, but no antitumor activity in others, with likewise variable influence on MYCN protein levels (8, 9). In addition, although there has been some clinical efficacy in early adult trials, most notably in NUT midline carcinoma with a canonical BRD4–NUT fusion oncoprotein (10), objective response rates have been low and mostly transient in other diseases (11, 12) and it has become clear that tumor cells can adapt in ways that restore MYC despite continued BET inhibition (13, 14).

Another illustrative example is the interaction between Aurora kinase A (AURKA) and MYCN and the effects of AURKA inhibitors. Early preclinical testing showed excellent activity of the AURKA inhibitor alisertib (MLN8237) in pediatric solid tumors (15), though this was independent of MYCN status. In addition, MYC-driven tumors were also sensitive to AURKA inhibition (16). Subsequent clinical testing in neuroblastoma demonstrated substantial toxicity and largely disappointing responses (17). However, AURKA binds to MYCN and sequesters it from degradation independent of its kinase activity (18). This raises the possibility that targeting this scaffolding function of AURKA may be more effective and more specific for MYCN, as the AURKA-binding domain of MYCN is poorly conserved with MYC (19) and a similar interaction between AURKA and MYC has not been as well characterized (20). Small molecules have been identified that bind to AURKA and alter its conformation in a way that prevents binding to MYCN and result in rapid MYCN degradation (21). More recently, a chemical degrader approach has also been taken (22). Although targeting the synthetic lethal interaction between MYCN and the kinase activity of AURKA has to date failed to provide an efficacious and specific therapeutic, it remains possible that targeting the MYCN-stabilizing function of the AURKA–MYCN complex will prove more successful. These and other examples support the hypothesis that sustained and specific inhibition of MYCN activity will require direct targeting of the deregulated protein or the MYCN complex.

Barriers to Direct and Specific Inhibition

Directly targeting MYCN poses substantial challenges that can be generalized to many transcription factors, but also some challenges that are unique to MYCN. Like many transcription factors, MYCN functions in the nucleus, has no known enzymatic function, and mediates its effects in the context of several multiprotein complexes that involve numerous protein–protein interactions and protein–DNA interactions. The protein–protein interaction surfaces in particular tend to be large and lack the defined hydrophobic pockets typically targeted by drug-like small molecules, and complex

formation tends to involve cooperation of multiple low-affinity interactions that are individually difficult to target.

MYCN also poses some unique challenges as a drug target. The N-terminal transcription-activating domains of MYC family proteins are intrinsically disordered in their monomeric forms (23, 24), and the C-terminal basic helix–loop–helix–leucine zipper (bHLH-LZ) domain lacks deep hydrophobic pockets (25). Certain N-terminal domains become structured in complex with binding partners, enabling limited structural studies of these domains (26, 27), but interactome studies have identified hundreds of interacting proteins (28, 29). Finally, MYCN is highly homologous to MYC and MYCL within the basic helix–loop–helix–leucine zipper (bHLH-LZ) domain and the 5 MYC boxes that have been shown to mediate much of MYC family protein function (2, 28, 30). Although there are some clear functional differences between MYC and MYCN, particularly with regard to their respective interactions with MIZ1 (31), it remains a conceptual challenge to inhibit the oncogenic function of MYCN while preserving the physiologic functions of MYC, including those in normal cell division and in wound healing.

Although these challenges remain formidable, a clearly appealing approach (outlined in Fig. 1) would be to identify small molecules that bind to MYCN in complex with an essential and specific binding partner, using structural information to guide drug design and/or optimization, and then link the small molecule to an E3 ligase binder to induce MYCN degradation. Here, we discuss the prior difficulties with such an approach and the scientific and technological advances that may now make it feasible. These concepts were discussed at a meeting in November 2019 that brought together experts in pediatric cancer, MYC and MYCN biology, structural biology, biochemistry, medicinal chemistry, and protein degradation technology to address the goal of directly targeting MYCN.

Targeting MYCN in Its Oncogenic Context

Modern target-based drug discovery has relied on identifying the structure of fragments or lead molecules bound to their target protein, which then allows for the rational optimization of the molecule to improve potency while preserving and/or improving physicochemical drug-like properties. This process cannot be applied to intrinsically disordered proteins such as MYC-family proteins in their monomeric form. Although there have been efforts—and some progress—in targeting the intrinsically disordered state of MYC (23, 32), the compounds developed to date have generally suffered from low potency and have not yet been turned successfully into credible lead drug compounds. However, it has long been appreciated that MYC-family proteins require binding partners to exert their tumorigenic function, and individual domains of MYC-family proteins have been demonstrated to form stable structures when complexed with interacting proteins, most prominently the bHLH-LZ domain of MYC in complex with MAX (25). Although the structure of MYCN in complex with MAX has not been solved itself, it is thought to be highly homologous to the MYC–MAX complex. Indeed, MYC–MAX disrupters have been identified and these also disrupt the MYCN–MAX interaction, further supporting the concept that MYC family proteins interact with MAX in a highly similar fashion (33). The structure of other domains of MYC-family proteins have been solved in complex with other interacting proteins, including KPNA1 (importin- α), BIN1, WDR5, TBP, and AURKA (19, 24, 26, 27, 34–36). These examples clearly demonstrate that MYC-family proteins can assume

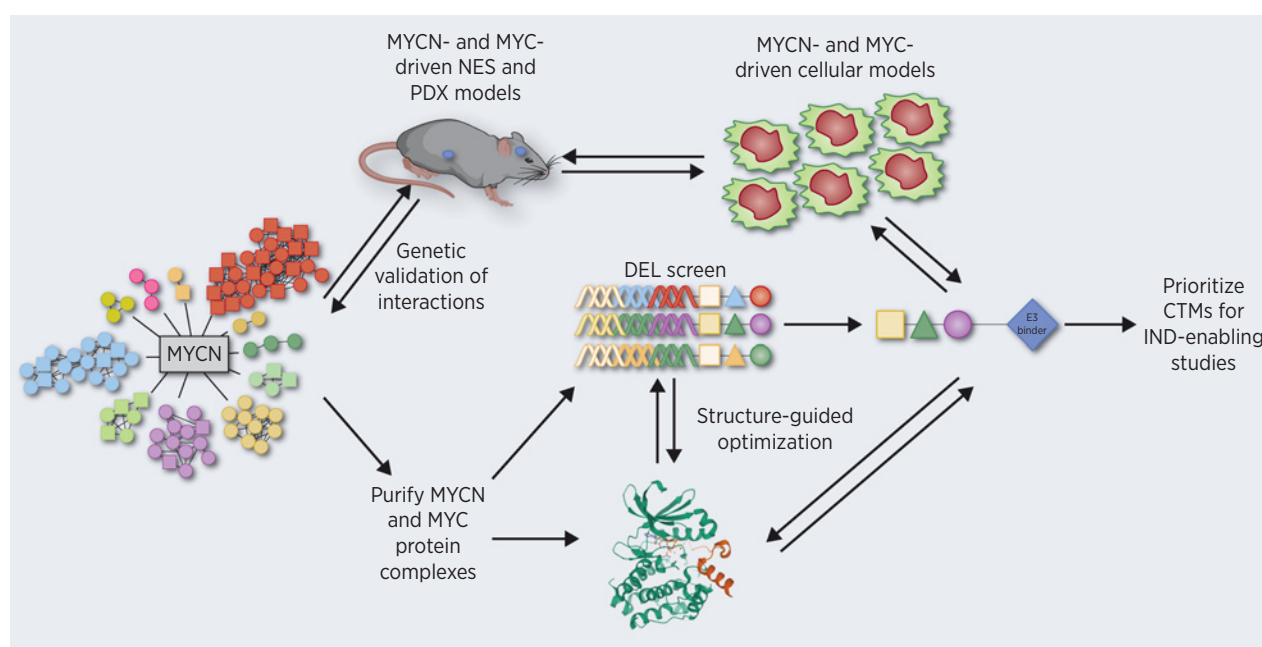


Figure 1.

Highlighted challenges and technological advances in identification and validation of a direct MYCN-targeting compound. Biochemically derived MYCN interactomes have recently revealed a large number of potential complexes to target and are contrasted with MYC interactomes (not shown). Which complexes are essential to tumor maintenance can now be identified using genetic loss-of-function testing in appropriate *in vivo* models [e.g., neural epithelial stem (NES) cell and PDX models]. Once oncogenic complexes are identified, these can be purified and used in structural studies, aided by advancements in cryo-EM technology. Challenges to the identification of small-molecule inhibitors can be addressed by using a bind and degrade approach. DNA-encoded chemical libraries (DEL) screens can facilitate the identification of binders unique to MYCN complexes as compared with MYC complexes. These compounds can then be linked to E3 ligase ligands to create proteolysis-targeting chimeras (PROTAC), optimized using structure as a guide, and tested in cell line and mouse models to confirm degradation, ensure antitumor activity, and determine selectivity. Once active and selective PROTACs are identified, they can be prioritized for IND-enabling studies and eventual clinical testing. Interactome modified from <http://pennlab.ca/research/#>, protein structure from the PDB (<https://www.rcsb.org/structure/5G1X>), and drawing tools from motifolio.com.

ordered states in the context of multiprotein complexes, providing a potential avenue for applying the tools of modern structure-based drug discovery.

Although targeting MYCN in a structured complex with an interacting protein has clear appeal, determining which interacting protein(s) to choose is challenging. The identity of the full complement of MYCN-interacting proteins has only recently been catalogued. This has not allowed for a substantial narrowing of focus, however, as interactome profiling has identified hundreds of proteins that can complex with MYC-family proteins (28, 29). This large number strongly suggests that there is not a single MYCN complex, but rather a number of different complexes that may differ in function and in their contribution to tumorigenesis and tumor maintenance and that change in composition throughout the cell cycle and across cell types or tumor types. Comparison between MYC and MYCN-binding proteins has revealed large numbers of common interactors, but also many MYCN-unique interactors.

Before undertaking laborious structural biology and drug-discovery campaigns against a given target, the MYCN-interacting protein complex should first be demonstrated to be essential for tumor maintenance in an appropriate *in vivo* model and not required for survival of normal cells. Such models and the appropriate genetic manipulation tools have only recently become available. For example, human neuroepithelial stem (NES) cell-based models provide one potentially appealing approach for this type of validation (37). NES

cells can be stably cultured and are amenable to gene editing technology. MYCN expression can be introduced, which results in medulloblastoma after orthotopic implantation (38), and similar models of neuroblastoma are also under development. A moderate number of interacting proteins can now be screened via genetic loss-of-function approaches to prioritize those that (when lost) result in tumor regression in MYCN-driven tumors but do not impede the growth of analogous MYC-driven tumors or of untransformed NES cells. Although alternative models could be used for validation of the centrality of a given interacting protein to MYCN's oncogenic function, this type of genetic validation is essential before initiating further drug discovery efforts and appropriate models are now available or will soon be available.

Limitations to Obtaining Structures of MYCN in Complex

Although structures of individual domains of MYC proteins have been identified in complex with interacting proteins (see discussion above), limitations in structural biology have made it difficult to obtain more extensive structural information about MYC complexes. All of the three central techniques used in structural biology [X-ray crystallography, nuclear magnetic resonance (NMR), and cryo-electron microscopy (cryo-EM)] are likely to have value and play complementary roles in obtaining the structural information about MYCN complexes needed to aid drug development. However, recent advances

in the ability to use cryo-EM to obtain structures of complexes of a variety of sizes is particularly important and is likely to make obtaining conformational data on oncogenic MYCN interactions more feasible.

Two parallel approaches could be undertaken to obtain the structure of purified MYCN complexes. The first, similar to what has been done previously (26), including with AURKA (19), is to map interaction domains using either NMR or cross-linking and mass spectroscopy, and then to use this information to pursue crystallographic determination of substructures of the MYCN complex. The advantages of this approach include identification of physiologically relevant MYCN interactions that might be of lower affinity, and a possibility of obtaining high-resolution structures of such complexes. However, an interaction requiring a large portion of MYCN and/or requiring multiple interactors may be difficult to probe using crystallography. Cryo-EM offers an alternative that can provide Ångström-level resolution of multiprotein complexes that are unlikely to crystallize in their full-length form, such as transcriptional complexes (39). These techniques can be used in a complementary fashion, with cryo-EM contributing an overall structure of the complex and X-ray crystallography or NMR focusing on smaller stable subcomplexes. Structural insights from these complementary techniques can then be incorporated to provide higher-resolution views of individual side chains in domains that may be less well visualized by cryo-EM. Lower-affinity interactions can be stabilized by cross-linking of nearby residues. These different modalities have been useful in the examination of complexes involving intrinsically disordered proteins like MYC, for example with the FACT complex (Facilitates Chromatin Transcription; refs. 40, 41). In addition, cryo-EM has successfully identified structures of such proteins when they form ordered structures in complex (42), though NMR has been applied more widely.

In addition to playing a role in the determination of structures of known MYCN complexes, advances in cryo-EM may also offer an opportunity to identify novel complexes. Graphene oxide (GO) covered cryo-EM functionalized with affinity tags (43) can be used to purify transient/lower-affinity MYCN complexes from tumors or cell lysates directly on the cryo-EM grid. In addition to providing structural information simultaneously on multiple different MYCN complexes, computational advances in cryo-EM analysis can potentially allow for the determination of previously unidentified proteins in these complexes (44).

Although multiple approaches could be pursued in parallel, the ultimate goal should be both to solve atomic resolution structures of oncogenic MYCN complexes that will be suitable for structure-based drug development and to obtain purified protein complexes that can be used for inhibitor screens. Given the described technological advances, this is now a much more achievable goal.

Challenges in Inhibitor Screening and Advances in Degradation Technology

Identifying small molecules that bind to transcription factors like MYC proteins is a substantial hurdle. Even if an effective binder is identified, a traditional therapeutic requires that the compound also interferes with protein function, a formidable challenge for a protein lacking enzymatic activity and functioning through protein–protein and protein–DNA interactions. However, the advent of proteolysis-targeting chimeras (PROTAC; also referred to as chimeric-targeting molecules; ref. 45) over the past 5 years has rendered “undruggable” targets such as MYCN potentially druggable. Rather than requiring a specific inhibitor of MYCN’s function, it may be sufficient to identify a MYCN-specific or MYCN complex-specific binder that can be linked

to an E3 ubiquitin ligase ligand to drive the rapid ubiquitination and degradation of MYCN. In addition to requiring only a binder (and not an inhibitor), this approach also may allow for enhanced specificity through choice of the E3 ligase that is engaged by the PROTAC. For example, tissue-specific expression of E3 ligases has been described previously (46), hence, recruiting an E3 ligase that is only expressed in tumor cells or neural cells could enhance tumor-specific activity and therapeutic index. Alternatively, by engaging only an E3 ligase with expression limited to MYCN-high cells, a molecule that binds both MYC and MYCN could be made into a *de facto* MYCN-specific degrader. One potential drawback of a degrader approach is that downregulation of the E3 ligase provides an additional potential resistance mechanism. For this reason, it is important to choose an E3 that is essential to tumor maintenance or to simultaneously apply ligands for multiple different ligases.

Importantly, adopting a degradation approach shifts the challenge in MYCN targeting to the identification of a molecule that specifically binds to MYCN or the MYCN oncogenic complex, with little or no binding to MYC or MYC complexes. Advances in small-molecule screening technology have also made this challenge easier to address. Fragment-based screening can identify low-affinity binders that can then be evolved into high-affinity lead compounds using structure-guided compound “growing” or by linking fragments together (47). If high-quality structures are available, advances in computational docking can identify synthetically accessible potential binders (48). Particularly promising is the advent of DNA-encoded chemical libraries (DEL) that have allowed hundreds of millions to billions of drug-like compounds to be rapidly screened for affinity against proteins or protein complexes of interest (49). Ideally, a systematic approach could be taken to use DELs to (i) identify binders to MYCN and several essential MYCN complexes validated *in vivo* as critical for tumorigenic functions of MYCN, followed by (ii) hit resynthesis and binding validation, then (iii) linkage to a number of different E3 ligase ligands, and (iv) analysis of the effect of the compounds on MYCN protein or complex stability. Such a comprehensive screening and follow-up campaign may exceed the capacity of academic investigators, but there are companies that are well tooled to perform such experiments and may be willing to participate in novel types of public–private partnerships.

Once validated binders and degraders are identified, structures incorporating the MYCN complex, the compound, and the appropriate E3 ligase can be solved to rationally optimize both binding moieties as well as the linker, in addition to making modifications to enhance predicted pharmacologic properties.

Rigorous Validation and Characterization of Compound Activity and Molecular Diagnostic-Linked Early-Phase Clinical Trials

Even if putative inhibitors of MYCN or a MYCN oncogenic complex had been identified previously, properly modeling their efficacy and specificity would have been a substantial challenge. However, there are now available a range of genetically defined models, both *in vitro* and *in vivo*, to validate potential inhibitors or degraders. Cell lines that are functionally dependent upon MYCN or MYC can both be used to demonstrate MYCN-specific degradation and growth inhibition, with the caveat that changes in expression in tissue culture of the E3 and of competing E3 substrates may influence specificity. Ideally, compounds advanced to *in vivo* testing for efficacy should demonstrate nanomolar potency,

several-fold MYCN selectivity, and undergo pharmacokinetic testing, including blood–brain barrier penetration analysis. The latter is essential to understand how the drug might be used in patients with brain tumors, as well as neuroblastomas that can metastasize to the central nervous system.

For *in vivo* testing, both patient-derived xenograft models (PDX) and genetically engineered mouse models (GEMM) are now widely available for the relevant diseases. Extensive PDX models of neuroblastoma, medulloblastoma, and ATRTs (50–52) have the advantages of being human cells with human MYCN and MYCN interactors, of providing sufficient diversity to model genetic heterogeneity among MYCN-driven tumors, and of providing MYC-driven tumors that can be used as controls. Although in autochthonous neuroblastoma and medulloblastoma GEMMs MYCN-interacting proteins are murine, non-germline GEMMs using human cells enable models in which interacting proteins are human. The GEMMs also do not provide as much heterogeneity as PDXs. However, given the long-appreciated role of MYCN in suppressing antigen presentation and creating an immune-depleted tumor microenvironment (53–55), it is essential to profile alterations in immune interactions upon MYCN depletion to understand the possible engagement of the adaptive immune system and how this might be enhanced. Orthogonal preclinical development of drug candidates will be essential for prioritizing the optimal drug(s) for early-phase clinical trials.

Advances in molecular diagnostics should allow for more precise early-phase clinical trials. Patients can be selected that have tumors with clear hyperactive MYCN signaling, both through copy-number changes and transcriptional profiling. Response can be followed over time both through traditional imaging modalities as well as through detection of MYCN in cell-free circulating tumor DNA (56). Finally, patients on early-phase clinical trial typically have suffered multiple relapses and received extensive immunosuppressive therapy. If pre-clinical testing demonstrates that drug efficacy depends on intact immunity, it may be necessary to incorporate immune function criteria into early-phase trials.

Conclusions

Driver transcription factors such as MYCN historically represent “undruggable” targets. For MYCN, this has been due to limitations in

understanding the biochemistry and structural biology of MYCN complexes, the inability to model those complexes *in vivo*, and the difficulty in identifying small-molecule inhibitors of nonenzymatic proteins like MYCN. No single advance, but rather progress on all of these fronts suggests that it is time to revisit a direct targeting approach, particularly in light of the continued failures of indirect approaches to produce an effective therapeutic. Here, we describe how developments in MYCN biology, structural biology (especially cryo-EM), drug screening, modeling of pediatric cancers in mice, and PROTAC/degrader technology have made direct targeting of MYCN a practical and feasible goal, which we expect will produce an important new therapeutic for several devastating childhood tumors.

Authors' Disclosures

R. Bayliss reports grants from Cancer Research UK during the conduct of the study. M.A. Lemmon reports other compensation from PHILIP Inc. (stock options as member of scientific advisory board) outside the submitted work. M. Rape reports personal fees from Nurix Tx and Monte Rosa Tx outside the submitted work. M.F. Roussel reports grants from NCI CA-096832 and NCI CA-02165 and other funding from ALSAC of SJCRH (financial support) outside the submitted work. K.M. Shokat reports personal fees from EFFECTOR Therapeutics, Roche, ERASCA, Ikena, Janssen Pharmaceuticals, Kumquat Biosciences, Kura Oncology, Merck, Nextech, Revolution Medicines, Turning Point, Type6 Therapeutics, Venthera, Wellspring Biosciences, and Initial Therapeutics outside the submitted work, as well as a patent for Degradors of Myc pending and owned by University of California San Francisco. W.P. Tansey reports a patent for 10,501,466 issued and 10,160,763 issued. No disclosures were reported by the other authors.

The Editor-in-Chief, Chi V. Dang, is an author on this article. In keeping with the AACR's editorial policy, the peer review of this submission was managed by a senior member of *Cancer Research's* editorial team; a member of the AACR Publications Committee rendered the final decision concerning acceptability.

Acknowledgments

The meeting and this report were supported by a grant from the Alex's Lemonade Stand Foundation, and the meeting was held at their Childhood Cancer Data Laboratory in Philadelphia, Pennsylvania.

The costs of publication of this article were defrayed in part by the payment of page charges. This article must therefore be hereby marked *advertisement* in accordance with 18 U.S.C. Section 1734 solely to indicate this fact.

Received October 12, 2020; revised December 28, 2020; accepted January 22, 2021; published first January 28, 2021.

References

- Dang CV. MYC on the path to cancer. *Cell* 2012;149:22–35.
- Rickman DS, Schulte JH, Eilers M. The expanding world of N-MYC-driven tumors. *Cancer Discov* 2018;8:150–63.
- Weiss WA, Aldape K, Mohapatra G, Feuerstein BG, Bishop JM. Targeted expression of MYCN causes neuroblastoma in transgenic mice. *EMBO J* 1997;16:2985–95.
- Swartling FJ, Grimmer MR, Hackett CS, Northcott PA, Fan QW, Goldenberg DD, et al. Pleiotropic role for MYCN in medulloblastoma. *Genes Dev* 2010;24:1059–72.
- Delmore JE, Issa GC, Lemieux ME, Rahl PB, Shi J, Jacobs HM, et al. BET bromodomain inhibition as a therapeutic strategy to target c-Myc. *Cell* 2011;146:904–17.
- Puissant A, Frumm SM, Alexe G, Bassil CF, Qi J, Chanthery YH, et al. Targeting MYCN in neuroblastoma by BET bromodomain inhibition. *Cancer Discov* 2013;3:308–23.
- Kato F, Fiorentino FP, Alibes A, Perucho M, Sanchez-Céspedes M, Kohno T, et al. MYCL is a target of a BET bromodomain inhibitor, JQ1, on growth suppression efficacy in small cell lung cancer cells. *Oncotarget* 2016;7:77378–88.
- Iniguez AB, Alexe G, Wang EJ, Roti G, Patel S, Chen L, et al. Resistance to epigenetic-targeted therapy engenders tumor cell vulnerabilities associated with enhancer remodeling. *Cancer Cell* 2018;34:922–38.
- Healy JR, Hart LS, Shazad AL, Gagliardi ME, Tsang M, Elias J, et al. Limited antitumor activity of combined BET and MEK inhibition in neuroblastoma. *Pediatr Blood Cancer* 2020;67:e28267.
- Stathis A, Zucca E, Bekradda M, Gomez-Roca C, Delord JP, de La Motte Rouge T, et al. Clinical response of carcinomas harboring the BRD4-NUT oncoprotein to the targeted bromodomain inhibitor OTX015/MK-8628. *Cancer Discov* 2016;6:492–500.
- Amorim S, Stathis A, Gleeson M, Iyengar S, Magarotto V, Leleu X, et al. Bromodomain inhibitor OTX015 in patients with lymphoma or multiple myeloma: a dose-escalation, open-label, pharmacokinetic, phase 1 study. *Lancet Haematol* 2016;3:e196–204.
- Lewin J, Soria JC, Stathis A, Delord JP, Peters S, Awada A, et al. Phase Ib trial with birabresib, a small-molecule inhibitor of bromodomain and extraterminal proteins, in patients with selected advanced solid tumors. *J Clin Oncol* 2018;36:3007–14.
- Rathert P, Roth M, Neumann T, Muerdter F, Roe JS, Muhan M, et al. Transcriptional plasticity promotes primary and acquired resistance to BET inhibition. *Nature* 2015;525:543–7.
- Fong CY, Gilan O, Lam EY, Rubin AF, Ftouni S, Tyler D, et al. BET inhibitor resistance emerges from leukaemia stem cells. *Nature* 2015;525:538–42.

15. Maris JM, Morton CL, Gorlick R, Kolb EA, Lock R, Carol H, et al. Initial testing of the aurora kinase a inhibitor MLN8237 by the pediatric preclinical testing program (PPTP). *Pediatr Blood Cancer* 2010;55:26–34.
16. Mollaoglu G, Guthrie MR, Bohm S, Bragelmann J, Can I, Ballieu PM, et al. MYC drives progression of small-cell lung cancer to a variant neuroendocrine subtype with vulnerability to aurora kinase inhibition. *Cancer Cell* 2017;31:270–85.
17. Mosse YP, Fox E, Teachey DT, Reid JM, Safgren SL, Carol H, et al. A phase II study of alisertib in children with recurrent/refractory solid tumors or leukemia: children's oncology group phase I and pilot consortium (ADV0921). *Clin Cancer Res* 2019;25:3229–38.
18. Otto T, Horn S, Brockmann M, Eilers U, Schuttrumpf L, Popov N, et al. Stabilization of N-Myc is a critical function of Aurora A in human neuroblastoma. *Cancer Cell* 2009;15:67–78.
19. Richards MW, Burgess SG, Poon E, Carstensen A, Eilers M, Chesler L, et al. Structural basis of N-Myc binding by Aurora-A and its destabilization by kinase inhibitors. *Proc Natl Acad Sci U S A* 2016;113:13726–31.
20. Dauch D, Rudalska R, Cossa G, Nault JC, Kang TW, Wuestefeld T, et al. A MYC-aurora kinase A protein complex represents an actionable drug target in p53-altered liver cancer. *Nat Med* 2016;22:744–53.
21. Gustafson WC, Meyerowitz JG, Nekritz EA, Chen J, Benes C, Charron E, et al. Drugging MYCN through an allosteric transition in Aurora kinase A. *Cancer Cell* 2014;26:414–27.
22. Adhikari B, Bozilovic J, Diebold M, Schwarz JD, Hofstetter J, Schroder M, et al. PROTAC-mediated degradation reveals a non-catalytic function of AURORA-A kinase. *Nat Chem Biol* 2020;16:1179–88.
23. Metallo SJ. Intrinsically disordered proteins are potential drug targets. *Curr Opin Chem Biol* 2010;14:481–8.
24. Andresen C, Helander S, Lemak A, Fares C, Csizmok V, Carlsson J, et al. Transient structure and dynamics in the disordered c-Myc transactivation domain affect Bin1 binding. *Nucleic Acids Res* 2012;40:6353–66.
25. Nair SK, Burley SK. X-ray structures of Myc-Max and Mad-Max recognizing DNA. Molecular bases of regulation by proto-oncogenic transcription factors. *Cell* 2003;112:193–205.
26. Bayliss R, Burgess SG, Leen E, Richards MW. A moving target: structure and disorder in pursuit of Myc inhibitors. *Biochem Soc Trans* 2017;45:709–17.
27. Wei Y, Resettec D, Li Z, Johansson-Akhe I, Ahlner A, Helander S, et al. Multiple direct interactions of TBP with the MYC oncoprotein. *Nat Struct Mol Biol* 2019;26:1035–43.
28. Kalkat M, Resettec D, Lourenco C, Chan PK, Wei Y, Shiah YJ, et al. MYC protein interactome profiling reveals functionally distinct regions that cooperate to drive tumorigenesis. *Mol Cell* 2018;72:836–48.
29. Buchel G, Carstensen A, Mak KY, Roeschert I, Leen E, Sumara O, et al. Association with aurora-a controls N-MYC-dependent promoter escape and pause release of RNA polymerase II during the cell cycle. *Cell Rep* 2017;21:3483–97.
30. Kohl NE, Legouy E, DePinho RA, Nisen PD, Smith RK, Gee CE, et al. Human N-myc is closely related in organization and nucleotide sequence to c-myc. *Nature* 1986;319:73–7.
31. Vo BT, Wolf E, Kawauchi D, Gebhardt A, Reh J, Finkelstein D, et al. The interaction of Myc with Miz1 defines medulloblastoma subgroup identity. *Cancer Cell* 2016;29:5–16.
32. Hammoudeh DI, Follis AV, Prochownik EV, Metallo SJ. Multiple independent binding sites for small-molecule inhibitors on the oncoprotein c-Myc. *J Am Chem Soc* 2009;131:7390–401.
33. Han H, Jain AD, Truica MI, Izquierdo-Ferrer J, Anker JF, Lysy B, et al. Small-molecule MYC inhibitors suppress tumor growth and enhance immunotherapy. *Cancer Cell* 2019;36:483–97.
34. Pineda-Lucena A, Ho CS, Mao DY, Sheng Y, Laister RC, Muhandiram R, et al. A structure-based model of the c-Myc/Bin1 protein interaction shows alternative splicing of Bin1 and c-Myc phosphorylation are key binding determinants. *J Mol Biol* 2005;351:182–94.
35. Conti E, Kuriyan J. Crystallographic analysis of the specific yet versatile recognition of distinct nuclear localization signals by karyopherin alpha. *Structure* 2000;8:329–38.
36. Thomas LR, Wang Q, Grieb BC, Phan J, Foshage AM, Sun Q, et al. Interaction with WDR5 promotes target gene recognition and tumorigenesis by MYC. *Mol Cell* 2015;58:440–52.
37. McLaren D, Gorba T, Marguerie de Rotrou A, Pillai G, Chappell C, Stacey A, et al. Automated large-scale culture and medium-throughput chemical screen for modulators of proliferation and viability of human induced pluripotent stem cell-derived neuroepithelial-like stem cells. *J Biomol Screen* 2013;18:258–68.
38. Huang M, Taylor J, Zhen Q, Gillmor AH, Miller ML, Weishaupt H, et al. Engineering genetic predisposition in human neuroepithelial stem cells recapitulates medulloblastoma tumorigenesis. *Cell Stem Cell* 2019;25:433–46.
39. Vos SM, Farnung L, Boehning M, Wigge C, Linden A, Urlaub H, et al. Structure of activated transcription complex Pol II-DSIF-PAF-SPT6. *Nature* 2018;560:607–12.
40. Liu Y, Zhou K, Zhang N, Wei H, Tan YZ, Zhang Z, et al. FACT caught in the act of manipulating the nucleosome. *Nature* 2020;577:426–31.
41. Tsunaka Y, Ohtomo H, Morikawa K, Nishimura Y. Partial replacement of nucleosomal DNA with human FACT induces dynamic exposure and acetylation of histone H3 N-terminal tails. *iScience* 2020;23:101641.
42. Gilman MSA, Liu C, Fung A, Behera I, Jordan P, Rigaux P, et al. Structure of the respiratory syncytial virus polymerase complex. *Cell* 2019;179:193–204.
43. Wang F, Liu Y, Yu Z, Li S, Feng S, Cheng Y, et al. General and robust covalently linked graphene oxide affinity grids for high-resolution cryo-EM. *Proc National Acad Sci U S A* 2020;117:24269–73.
44. Ho CM, Li X, Lai M, Terwilliger TC, Beck JR, Wohlschlegel J, et al. Bottom-up structural proteomics: cryoEM of protein complexes enriched from the cellular milieu. *Nat Methods* 2020;17:79–85.
45. Burslem GM, Crews CM. Proteolysis-targeting chimeras as therapeutics and tools for biological discovery. *Cell* 2020;181:102–14.
46. Kaneko M, Iwase I, Yamasaki Y, Takai T, Wu Y, Kanemoto S, et al. Genome-wide identification and gene expression profiling of ubiquitin ligases for endoplasmic reticulum protein degradation. *Sci Rep* 2016;6:30955.
47. Erlanson DA, Fesik SW, Hubbard RE, Jahnke W, Jhoti H. Twenty years on: the impact of fragments on drug discovery. *Nat Rev Drug Discov* 2016;15:605–19.
48. Lyu J, Wang S, Balias TE, Singh I, Levit A, Moroz YS, et al. Ultra-large library docking for discovering new chemotypes. *Nature* 2019;566:224–9.
49. Neri D, Lerner RA. DNA-encoded chemical libraries: a selection system based on endowing organic compounds with amplifiable information. *Annu Rev Biochem* 2018;87:479–502.
50. Stewart E, Federico SM, Chen X, Shelat AA, Bradley C, Gordon B, et al. Orthotopic patient-derived xenografts of paediatric solid tumours. *Nature* 2017;549:96–100.
51. Rokita JL, Rathi KS, Cardenas MF, Upton KA, Jayaseelan J, Cross KL, et al. Genomic profiling of childhood tumor patient-derived xenograft models to enable rational clinical trial design. *Cell Rep* 2019;29:1675–89.
52. Smith KS, Xu K, Mercer KS, Boop F, Klimo P, DeCupere M, et al. Patient-derived orthotopic xenografts of pediatric brain tumors: a St. Jude resource. *Acta Neuropathol* 2020;140:209–25.
53. Bernards R, Dessain SK, Weinberg RA. N-myc amplification causes down-modulation of MHC class I antigen expression in neuroblastoma. *Cell* 1986;47:667–74.
54. Layer JP, Kronmuller MT, Quast T, van den Boorn-Konijnenberg D, Effern M, Hinze D, et al. Amplification of N-Myc is associated with a T-cell-poor microenvironment in metastatic neuroblastoma restraining interferon pathway activity and chemokine expression. *Oncoimmunology* 2017;6:e1320626.
55. Wei JS, Kuznetsov IB, Zhang S, Song YK, Asgharzadeh S, Sindiri S, et al. Clinically relevant cytotoxic immune cell signatures and clonal expansion of T-cell receptors in high-risk MYCN-not-amplified human neuroblastoma. *Clin Cancer Res* 2018;24:5673–84.
56. Chicard M, Boyault S, Colmet Daage L, Richer W, Gentien D, Pierron G, et al. Genomic copy-number profiling using circulating free tumor DNA highlights heterogeneity in neuroblastoma. *Clin Cancer Res* 2016;22:5564–73.

Cancer Research

The Journal of Cancer Research (1916–1930) | The American Journal of Cancer (1931–1940)

Drugging the "Undruggable" MYCN Oncogenic Transcription Factor: Overcoming Previous Obstacles to Impact Childhood Cancers

Adam J. Wolpaw, Richard Bayliss, Gabriele Büchel, et al.

Cancer Res 2021;81:1627-1632. Published OnlineFirst January 28, 2021.

Updated version Access the most recent version of this article at:
doi:[10.1158/0008-5472.CAN-20-3108](https://doi.org/10.1158/0008-5472.CAN-20-3108)

Cited articles This article cites 56 articles, 11 of which you can access for free at:
<http://cancerres.aacrjournals.org/content/81/7/1627.full#ref-list-1>

E-mail alerts [Sign up to receive free email-alerts](#) related to this article or journal.

Reprints and Subscriptions To order reprints of this article or to subscribe to the journal, contact the AACR Publications Department at pubs@aacr.org.

Permissions To request permission to re-use all or part of this article, use this link
<http://cancerres.aacrjournals.org/content/81/7/1627>.
Click on "Request Permissions" which will take you to the Copyright Clearance Center's (CCC) Rightslink site.